

This are the design guidelines

Font : Calibri

Primary title font colour: #0070C0

#E97132

#0F9ED5

#196B24

- ☐ If you have bullet point make the box or dots blue
- ☐ Item two
- ☐ Item three

Blue Company

UAV in Search and Rescue

The first 72 hours are the most critical, as survival rates drop to 5–10% beyond this window, making immediate victim localization critical [3].

UAVs have become valuable tools in SAR missions

- ☐ Fast deployment
- ☐ Sensing capabilities
- ☐ Cost effective
- ☐ Access to hazardous places
- ☐ Realtime data collection



The Scenario

A **hiker** has gone **missing** in Fenswood **Wilderness**. A broad search region is defined, with the drone tasked to survey the area systematically and identify items of interest such as clothing and equipment. Mid-search, the hiker activates a Personal Locator Beacon (PLB), redirecting the mission to a focused area.

The Objective



Search the area



Identify the hiker



Deploy first aid kit

Hardware Components

Hexsoon EDU-450

Cube Orange FC

Here 3+ GPS

FrSky RC System

Tarot Release Mechanism

Raspberry Pi 5

Pi Global Shutter Camera

6mm Wide Angle Lens

First Aid Kit



Fry Sky



Deployment mechanism



First Aid Kit



Raspberry Pi 5 desktop

Untitled Map

Write a description for your map.

Legend

-  Feature 1
-  Feature 2
-  Flight Area
-  Focus Area (example0)
-  Focus Area (example1)
-  Focus Area (example2)
-  Focus Area (example3)
-  Focus Area (example4)
-  Long Ashton Community Centre
-  Peel Park
-  SSSI
-  Survey Area
-  Take-Off Location
-  The Bird in Hand
-  Untitled Path





Fenswood Farm
An overview of the SAR mission area

Mission Field & Environment

- Legend**
- Flight Area
 - Focus Area (example0)
 - Focus Area (example1)
 - Focus Area (example2)
 - Focus Area (example3)
 - Focus Area (example4)
 - SSSI
 - Survey Area
 - Take-Off Location



Fenswood Farm

An overview of the SAR mission area

- Legend**
- Flight Area
 - Focus Area (example0)
 - Focus Area (example1)
 - Focus Area (example2)
 - Focus Area (example3)
 - Focus Area (example4)
 - SSSI
 - Survey Area
 - Take-Off Location

Take-Off Location

PHENIX CO., Exclusive Hampers

Liane Baker Hair Studio

Perry House by C



Fenswood Farm

An overview of the SAR mission area

- Legend**
- Flight Area
 - Focus Area (example0)
 - Focus Area (example1)
 - Focus Area (example2)
 - Focus Area (example3)
 - Focus Area (example4)
 - SSSI
 - Survey Area
 - Take-Off Location



Fenswood Farm

An overview of the SAR mission area

- Legend**
- Flight Area
 - Focus Area (example0)
 - Focus Area (example1)
 - Focus Area (example2)
 - Focus Area (example3)
 - Focus Area (example4)
 - SSSI
 - Survey Area
 - Take-Off Location

Take-Off Location

EDEN & CO. (Eden Mill Farm)

Lane Baker Hair Studio

Perry House by Clift



Fenswood Farm

An overview of the SAR mission area

- Legend**
- Flight Area
 - Focus Area (example0)
 - Focus Area (example1)
 - Focus Area (example2)
 - Focus Area (example3)
 - Focus Area (example4)
 - SSSI
 - Survey Area
 - Take-Off Location

Take-Off Location

EDEN & CO. (Eden Farm)

Lane Baker Hair Studio

GSH

Perry House by Clift



Fenswood Farm
An overview of the SAR mission area

- Legend**
- Flight Area
 - Focus Area (example0)
 - Focus Area (example1)
 - Focus Area (example2)
 - Focus Area (example3)
 - Focus Area (example4)
 - SSSI
 - Survey Area
 - Take-Off Location



Fenswood Farm

An overview of the SAR mission area

Legend

Flight Area

Focus Area (example0)

Focus Area (example1)

Focus Area (example2)

Focus Area (example3)

Focus Area (example4)

SSSI

Survey Area

Take-Off Location



Fenswood Farm
An overview of the SAR mission area

- Legend**
- Flight Area
 - Focus Area (example0)
 - Focus Area (example1)
 - Focus Area (example2)
 - Focus Area (example3)
 - Focus Area (example4)
 - SSSI
 - Survey Area
 - Take-Off Location



Fenswood Farm

An overview of the SAR mission area

- Legend**
- Flight Area
 - Focus Area (example0)
 - Focus Area (example1)
 - Focus Area (example2)
 - Focus Area (example3)
 - Focus Area (example4)
 - SSSI
 - Survey Area
 - Take-Off Location



Fenswood Farm

An overview of the SAR mission area

- Legend**
- Flight Area
 - Focus Area (example0)
 - Focus Area (example1)
 - Focus Area (example2)
 - Focus Area (example3)
 - Focus Area (example4)
 - SSSI
 - Survey Area
 - Take-Off Location

Take-Off Location

Eden & Co. (Eden) Farm

Perry House by Clift



Fenswood Farm

An overview of the SAR mission area

Legend

Flight Area

Focus Area (example0)

Focus Area (example1)

Focus Area (example2)

Focus Area (example3)

Focus Area (example4)

SSSI

Survey Area

Take-Off Location



Fenswood Farm

An overview of the SAR mission area

- Legend**
- Flight Area
 - Focus Area (example0)
 - Focus Area (example1)
 - Focus Area (example2)
 - Focus Area (example3)
 - Focus Area (example4)
 - SSSI
 - Survey Area
 - Take-Off Location

Road



Take-Off Location

Railway Line



EDEN & CO. (Eden) Timbers
Lane Baker Hair Studio
GSH Architects
Perry House by Cliftonvalley Apartments

Fenswood Farm

An overview of the SAR mission area

Legend

 Flight Area

 Focus Area (example0)

 Focus Area (example1)

 Focus Area (example2)

 Focus Area (example3)

 Focus Area (example4)

 SSSI

 Survey Area

 Take-Off Location





Fenswood Farm

An overview of the SAR mission area

Legend

Flight Area

Focus Area (example0)

Focus Area (example1)

Focus Area (example2)

Focus Area (example3)

Focus Area (example4)

SSSI

Survey Area

Take-Off Location

Road

Take-Off Location

Railway Line

Footpath

Public Footpath

Fenswood Farm

An overview of the SAR mission area

Legend



 Flight Area



 Focus Area (example0)



 Focus Area (example1)



 Focus Area (example2)



 Focus Area (example3)



 Focus Area (example4)



 SSSI



 Survey Area



 Take-Off Location



Fenswood Farm

An overview of the SAR mission area

Legend

- Flight Area
- Focus Area (example0)
- Focus Area (example1)
- Focus Area (example2)
- Focus Area (example3)
- Focus Area (example4)
- SSSI
- Survey Area
- Take-Off Location



OVERVIEW

IDLE

PRE-AUTO
CHECK

SEARCH

REPLAN

FOCUS

DELIVER

SAFE RTL

SAFETY

SUMMARY

Mission Flow & Control Authority

State machine orchestrates the full SAR mission.
Pilot retains override authority at every stage.

Safety Pilot

- ❑ RC transmitter has hardware priority.
- ❑ Can override to LOITER at any time.
- ❑ Firmware-enforced, not software.

Operator

- ❑ Configures mission via GUI.
- ❑ Approves all autonomous transitions.
- ❑ Confirms detections & landing.

State Machine [SM]

- ❑ Executes uploaded missions in AUTO.
- ❑ Monitors telemetry & CV pipeline.
- ❑ Never fights pilot override.



GUIDED Gate: The state machine can **NEVER** request AUTO unless the pilot has physically placed the RC switch in GUIDED — explicit consent before every autonomous phase.



Pilot (RC)



Operator (GUI)



State Machine (AUTO)

OVERVIEW

IDLE

PRE-AUTO
CHECK

SEARCH

REPLAN

FOCUS

DELIVER

SAFE RTL

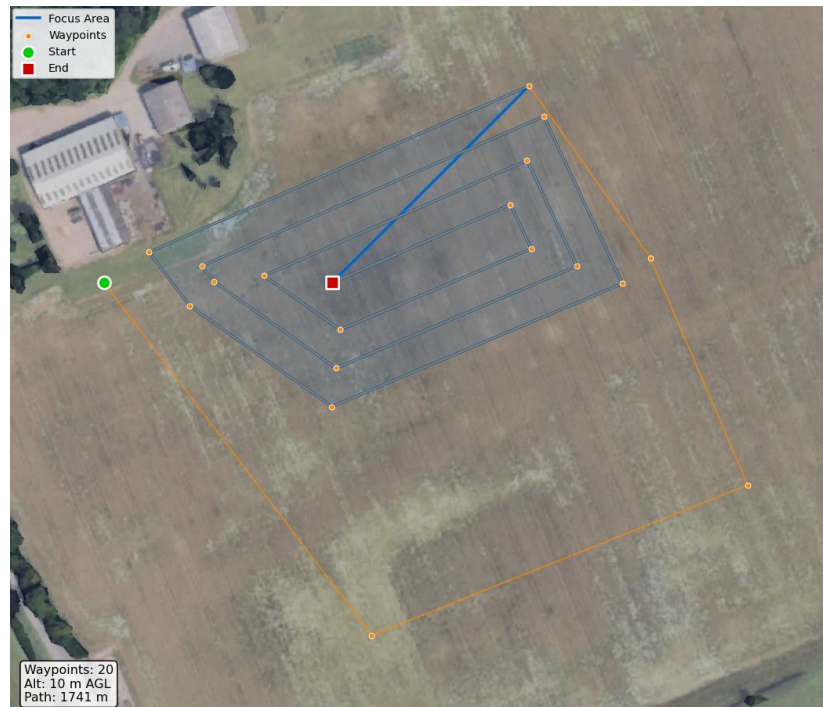
SAFETY

SUMMARY

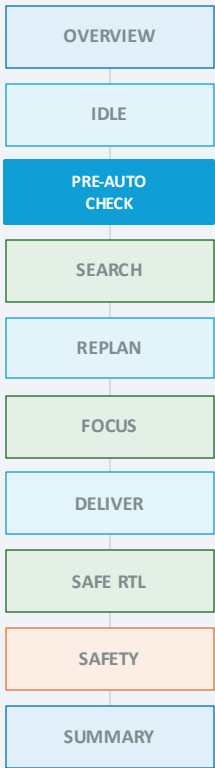
IDLE — Mission Setup

Operator configures mission via GUI. Drone arms and takes off to search altitude via GUIDED_TAKEOFF.

- 1 Upload Fences** — flight area (inclusion) + SSSI (exclusion) from KML
- 2 Generate Pattern** — lawnmower search + SSSI corridor transit waypoints
- 3 Preview & Confirm** — operator reviews pattern on map, confirms or cancels
- 4 Upload Mission** — validated waypoints sent to autopilot via MAVLink
- 5 Guided Take-off** — arms drone + climbs to search altitude in GUIDED
- 6 Open Checklist** — transitions to PRE_AUTO_CHECK



Pattern Preview



PRE-AUTO CHECK — Pre-Flight Gate

9 conditions must be green before autonomous flight is permitted.

Automated Checks (8)

- ✓ Heartbeat — consecutive miss count < limit
- ✓ GPS Fix — fix type $\geq 3D$
- ✓ GPS Satellites — count $\geq$ minimum
- ✓ Battery — percentage $\geq$ threshold
- ✓ Mode = GUIDED — pilot consent signal
- ✓ Armed — drone is armed
- ✓ Altitude — within tolerance of search alt
- ✓ Mission & Fences — uploaded to autopilot

Manual Check (1)

- ✓ Safety Pilot Ready — operator checkbox

PRE-AUTO CHECKLIST

- Heartbeat
- GPS quality
- Battery
- Mode = GUIDED
- Armed
- Altitude in range
- Mission uploaded
- Fences uploaded
- ☑ Safety pilot ready

START SEARCH ALL CHECKS PASSED

Key: Mode = GUIDED is the pilot's consent signal.
RC switch: LOITER / GUIDED / LOITER. Centre = 'I am handing over.'

OVERVIEW

IDLE

PRE-AUTO
CHECK

SEARCH

REPLAN

FOCUS

DELIVER

SAFE RTL

SAFETY

SUMMARY

SEARCH — Autonomous Flight

Inward spiral path navigation in AUTO mode. Pilot can override at any time.

Navigate to Search Area

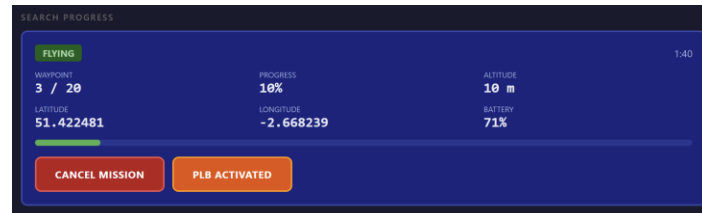
Request AUTO, then navigate pre-defined SSSI corridor at transit speed.

Monitor Autonomous Search

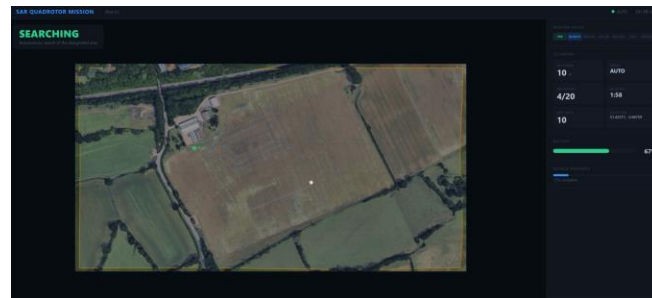
Navigate waypoints at initial search speed, while monitoring telemetry.

Operator and Spectator Display

Drone movement tracked and presented to spectator.
Waypoint completion tracking presented to operator and spectator.



Operator View



Spectator View

RC Override: Pilot flips to LOITER → SM stops all commands → GUI shows PAUSED → pilot returns to GUIDED → SM requests AUTO → resumes from current WP.

OVERVIEW

IDLE

PRE-AUTO
CHECK

SEARCH

REPLAN

FOCUS

DELIVER

SAFE RTL

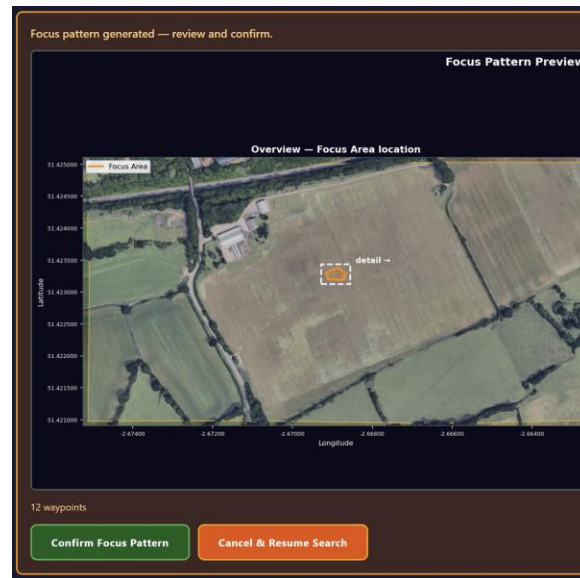
SAFETY

SUMMARY

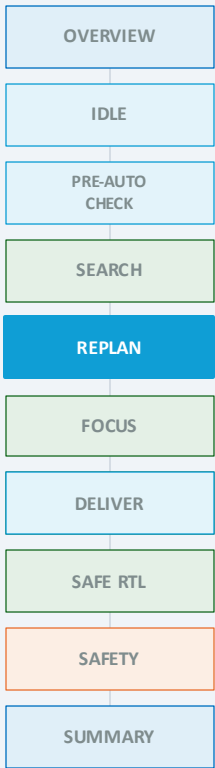
REPLAN — PLB Response

Drone holds in GUIDED while operator loads PLB focus area and reviews new pattern.

- 1 **SEARCH detects PLB button** → switches to GUIDED → drone holds position
- 2 **Operator drops Focus Area KML** and presses 'Load PLB KML' on dashboard
- 3 **System parses KML**, generates tighter focus lawnmower, renders preview
- 4 **Operator reviews preview** — confirm or cancel
- 5 **On confirm:** re-upload fences (base + focus inclusion), upload new mission



Focus Pattern Preview

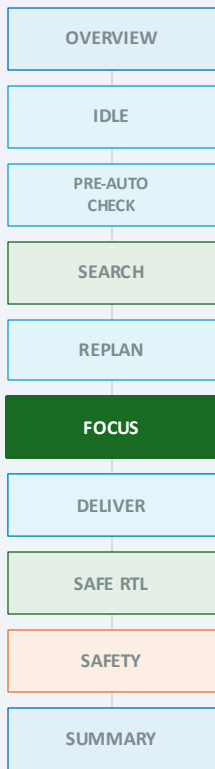


REPLAN — PLB Response

Drone holds in GUIDED while operator loads PLB focus area and reviews new pattern.

- 1 **SEARCH detects PLB button** → switches to GUIDED → drone holds position
- 2 **Operator drops Focus Area KML** and presses 'Load PLB KML' on dashboard
- 3 **System parses KML**, generates tighter focus lawnmower, renders preview
- 4 **Operator reviews preview** — confirm or cancel
- 5 **On confirm:** re-upload fences (base + focus inclusion), upload new mission



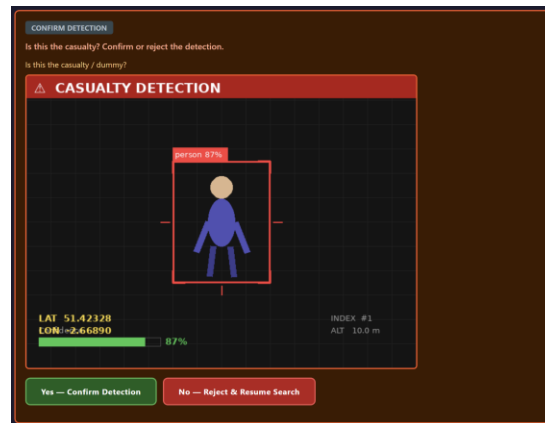


FOCUS — Tighter Search

AUTO mode over the PLB focus area. CV pipeline scans for casualties.

How It Works

- ☐ Tighter lawnmower pattern over PLB polygon.
- ☐ AUTO mode waypoint navigation.
- ☐ Reduced speed for focussed search.
- ☐ Telemetry monitored and GUIs updated accordingly.
- ☐ CV pipeline drops geotagged images into detection directory.
- ☐ State machine monitors for new files.
- ☐ On detection, operator confirms or rejects.



Detection Output

On Detection: CV pipeline writes geotagged image → state machine detects new file → operator sees detection card on GUI with image → operator confirms or dismisses → confirmed detection triggers transition to DELIVER.

Pattern Complete, No Detection: Focus lawnmower finishes with no CV hit → state machine returns to SEARCH to resume from current WP.

OVERVIEW

IDLE

PRE-AUTO
CHECK

SEARCH

REPLAN

FOCUS

DELIVER

SAFE RTL

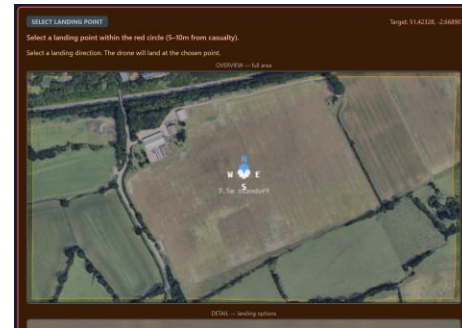
SAFETY

SUMMARY

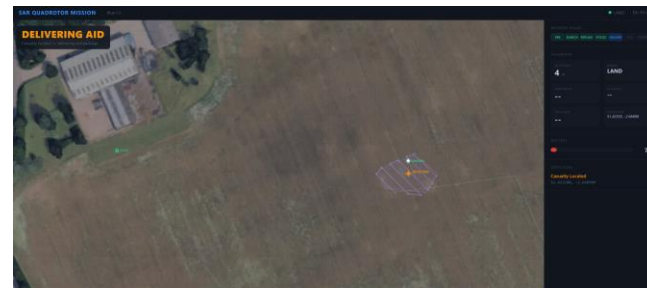
DELIVER — First-Aid Delivery

Operator confirms detection, selects landing point, and the drone delivers autonomously.

- 1 **Geotag Uploaded** — CV system uploads new image with aggregated geotag of dummy's location
- 2 **Select landing point** — operator selects landing target on interactive preview map
- 3 **Slow approach** — drone flies to landing point at 1 m/s
- 4 **Land** — drone lands at selected point, motors disarm
- 5 **Deploy payload** — operator confirms first-aid kit has been released
- 6 **Relaunch** — drone re-arms, climbs to search altitude → transitions to SAFE_RTL

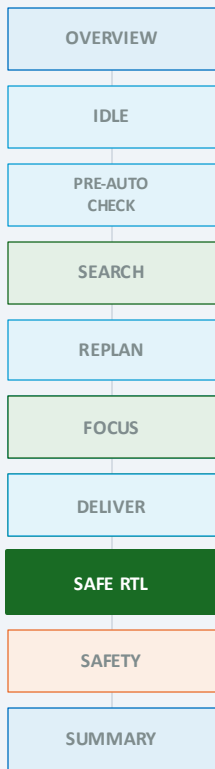


Landing Selection



Approach & Land

Exits to: **SAFE_RTL** (deploy → relaunch → corridor home) | **IDLE** (cancel at any phase)



SAFE RTL — Return to Launch

After delivery, the drone flies home via the SSSI corridor and lands at the original launch point.



Home Point Saved in State Machine from Initial Take-off Location

ArduPilot updates its internal HOME position every time the drone disarms and re-arms. During DELIVER the drone lands, deploys the first-aid kit, then re-arms and relaunches. This updates the ArduPilot's HOME to the casualty's location.

The state machine captures the original launch GPS coordinates at startup and uses that saved home point for the return-to-launch waypoint, bypassing ArduPilot's updated HOME entirely.

Return Sequence

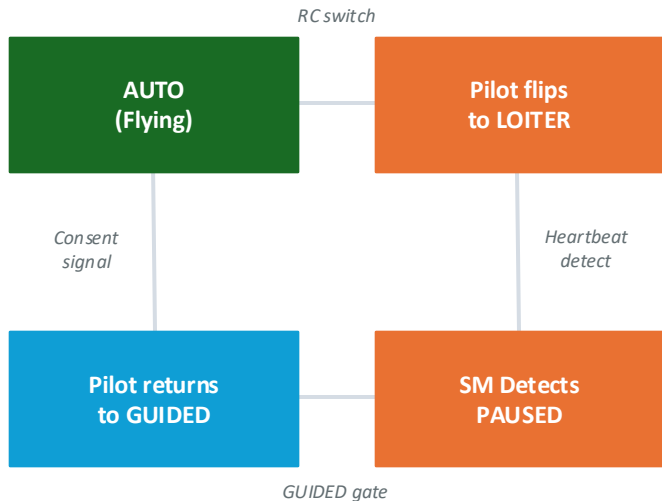
- 1 **Relaunch from casualty site** — climb to search altitude in GUIDED
- 2 **Upload corridor-home mission** — SSSI transit waypoints + saved home point as final WP
- 3 **GUIDED gate** — transmitter set to GUIDED → SM requests AUTO
- 4 **Fly corridor home** — AUTO mode along SSSI transit route to original launch point
- 5 **Land & disarm** — drone lands at saved home, state machine transitions to IDLE

Exits to: **IDLE** (landed & disarmed)

OVERVIEW
IDLE
PRE-AUTO CHECK
SEARCH
REPLAN
FOCUS
DELIVER
SAFE RTL
SAFETY
SUMMARY

Safety Architecture: Override & Recovery

Override & Resume Cycle



Safety Guarantees

1. Hardware Priority

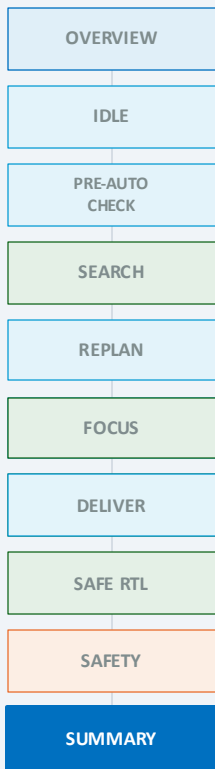
ArduPilot gives the RC transmitter unconditional priority over MAVLink. No software command can override a physical stick input.

2. GUIDED Consent Gate

The state machine will NEVER request AUTO unless the flight controller reports GUIDED — the pilot must physically set this.

3. No Fight-Back

On override, the state machine stops all commands, enters PAUSED, and waits. No re-entry to AUTO without fresh GUIDED consent.



State Machine Summary

Human-in-the-Loop at every stage

No autonomous flight without operator setup, checklist approval, and pilot GUIDED consent. Every detection, landing point, and deploy requires operator confirmation.

Pilot override is unconditional

RC has firmware-level priority. The state machine detects override via heartbeat, stops all commands, and waits for fresh GUIDED consent before resuming. No fight-back logic exists.

Designed for the spectator

The operator GUI and spectator dashboard give real-time visibility into state, waypoint progress, battery, altitude, and mission status — so you can follow every phase of the mission as it happens.

Ready for demo.

Data Collection and Vision Training

Synthetic Data Generation

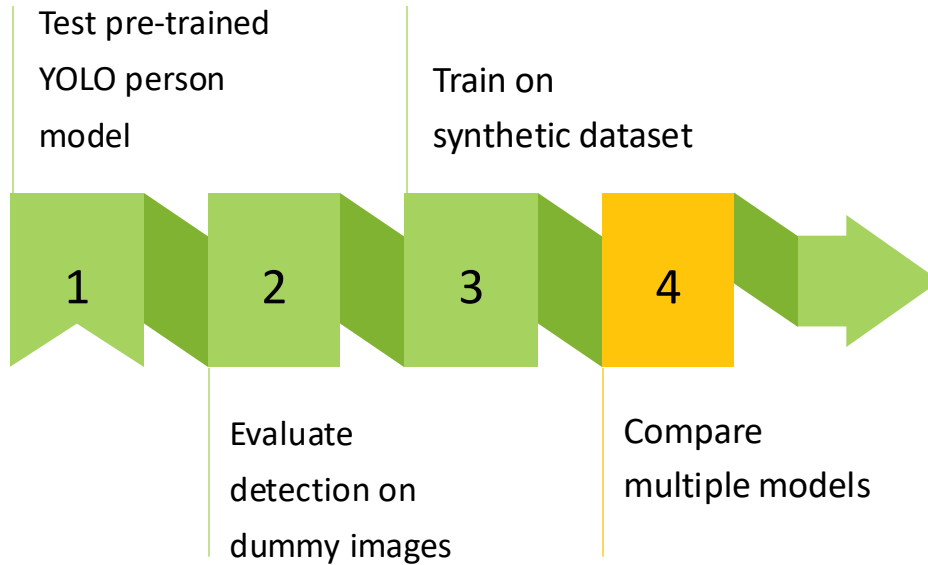
- 01 Multiple dummy poses (lying, side, curled)
- 02 Different colour variations (clothes, high-visibility)
- 03 Diverse backgrounds (grass, soil, rocks)
- 04 Lighting variation (brightness, contrast)
- 05 Partial occlusion (leaves, grass)

No real dummy dataset
available at early stage

Total: Thousands of
images generated

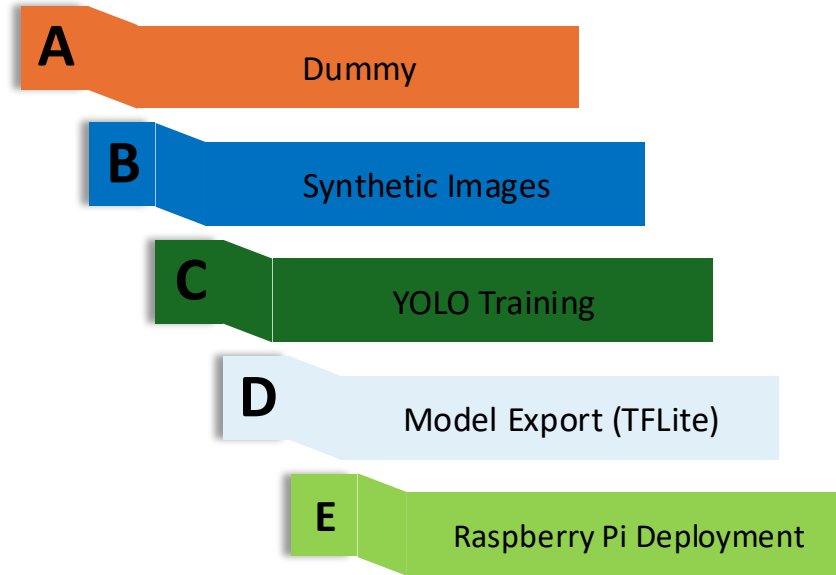


Model Training Strategy



- YOLO: one stage real time; pretrained person; simplest Pi deployment.
- Synthetic data: limited real dummy photos; vary pose, occlusion, lighting, backgrounds.
- Compare variants: accuracy latency trade off; choose best mean average precision plus frames per second.
- TensorFlow Lite on Pi: verify inference speed, memory, and stability constraints.

Vision Training and Deployment Pipeline



- Synthetic data helps generalisation only if variation is realistic.
- Model selection must consider latency and memory, not only metrics.

CV to Target Estimate Pipeline

Camera Capture

1456 × 1088 px
IMX296 global shutter

YOLOv8n Inference

NCNN backend — 73 ms
640×640 input tensor

Pixel → GPS

FOV projection +
altitude
Calibrated focal length

Attitude Correction

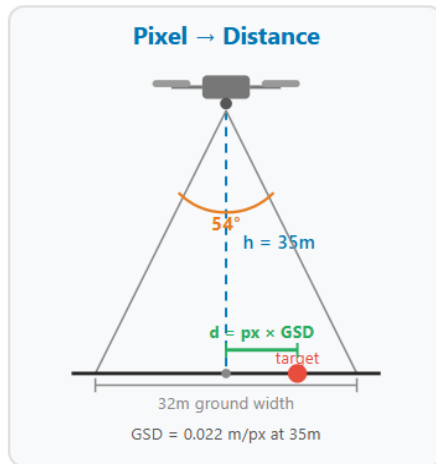
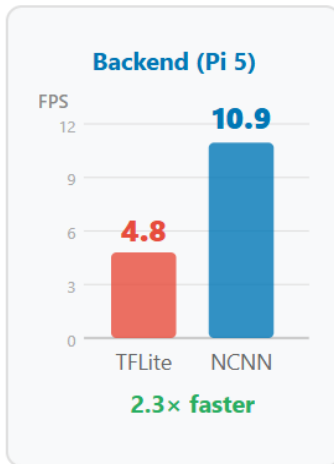
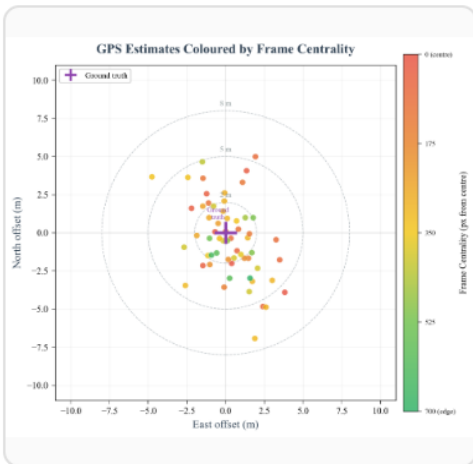
Pitch + roll ray-trace
From flight controller



Target GPS

~9 FPS effective
CEP50 = 2.2 m

Camera & Calibration



IMX296 Global Shutter

1456 × 1088 px native resolution, 30 fps capture

Global shutter — no rolling-shutter distortion at any speed

Sub-pixel motion blur (<0.5 px at 8 m/s) — no gimbal needed

Pixel → Real-World Distance

FOV calibrated — focal length 5.46mm (22% off datasheet!)

$\text{GSD} = 0.022 \text{ m/px at } 35\text{m altitude}$

Ground footprint = 32 × 24 m at 35m alt

Pixel offset × GSD → metre offset → add to drone GPS

Lens Distortion Correction

Checkerboard calibration — 14×9 board, RMS = 0.4 px

Precomputed undistortion maps applied every frame

Cost: +1.5ms per frame (negligible on 206ms inference)

Simulation and Sim2Real Ladder

Simulation Recording
takeoff → search → detect → centre → verify → land

Sim2Real Ladder

(never fly something you haven't proven at every level below it)



Real Mission

Full autonomous SAR — demo day

GOAL

FLIGHT MISSIONS

- ☐ M4 Auto detect + hover
- ☒ M3 Waypoint flight
- ☒ M2 Passive flight
- ☒ M1 Bench integration

GROUND VERIFICATION

- ☒ G2 Outdoor calibration
- ☒ G1 Unit tests + dry-run

SIMULATION FOUNDATION

All 12 mission requirements verified end-to-end in SITL

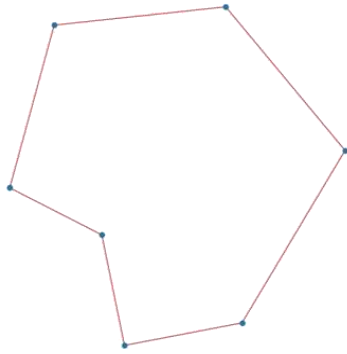
Spiral search pattern	Attitude-compensated GPS
AI detection + centering	Payload deploy sequence
Geofence + SSSI no-fly zone	RTL failsafes
PLB beacon redirect	Offset landing

Simulated Realism

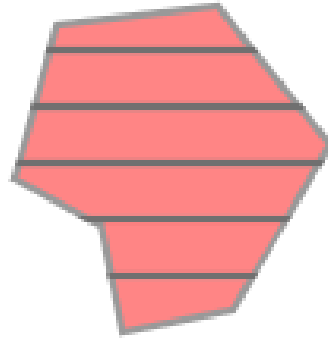
Same 1456×1088 resolution as real camera
Camera tilt simulation (pitch/roll)
Motion blur at flight speed
GPS drift noise ($\pm 3\text{m}$)
Altitude-correct target scaling
Synthetic targets on real backgrounds

"Not just waypoints — full state machine with realistic camera, detection, and GPS behavior"

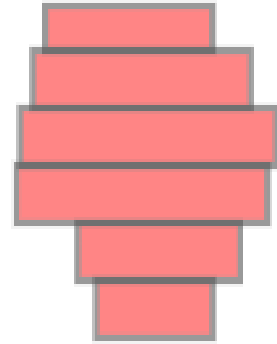
Step 1: Defining the PLB Region



Polygon Created



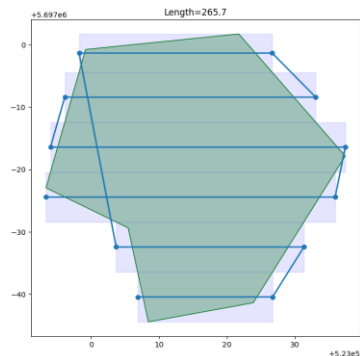
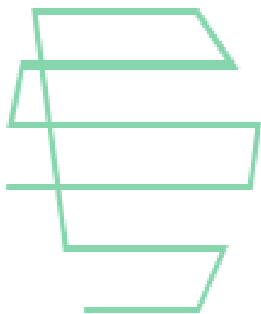
**Polygon divided into
strips**



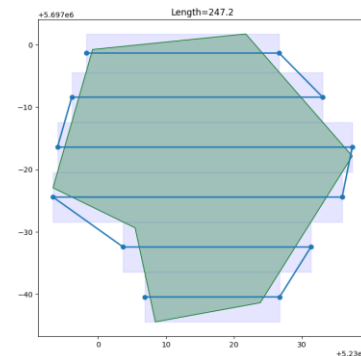
**Rectangular coverage
strips expanded**

Step 2: Simple vs. Greedy Path

Simple Path

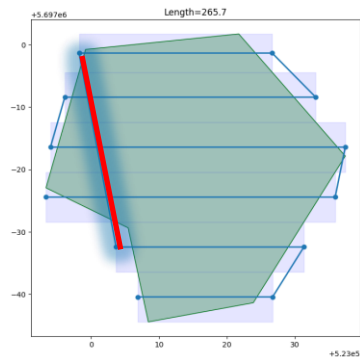
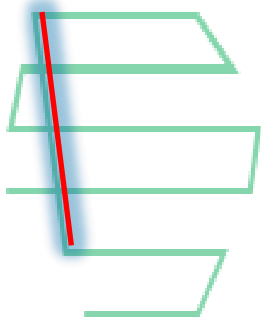


Greedy Path

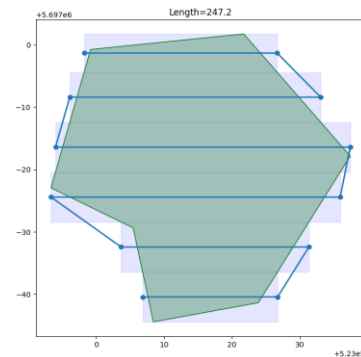


Step 2: Simple vs. Greedy Path

Simple Path

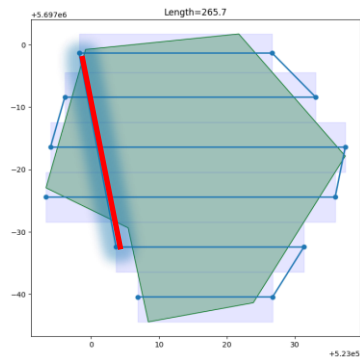
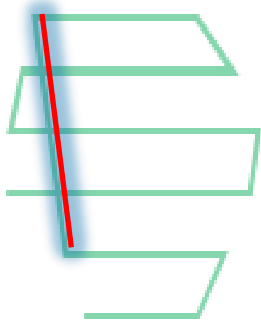


Greedy Path



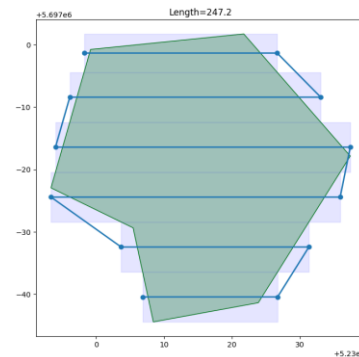
Step 2: Simple vs. Greedy Path

Simple Path



Length travelled = 265.7 meters

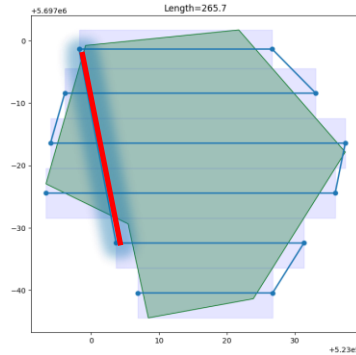
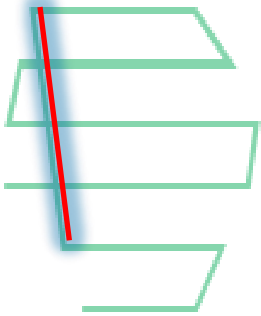
Greedy Path



Length travelled = 247.2 meters

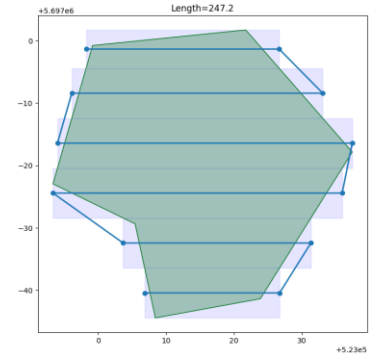
Step 2: Simple vs. Greedy Path

Simple Path

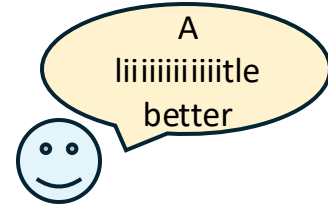


Length travelled = 265.7 meters

Greedy Path

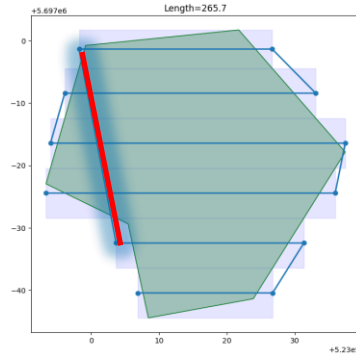
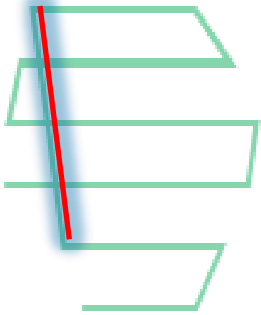


Length travelled = 247.2 meters



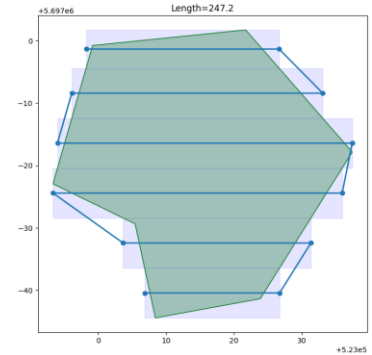
Step 2: Simple vs. Greedy Path

Simple Path



Length travelled = 265.7 meters

Greedy Path



Length travelled = 247.2 meters



A
little
better

Path Planning Contribution

Key Innovation: Introduce a modular `PATH_PLANNER` sub-module.

- ✓ Simulation, comparison and detailed data test on 2 path planning algorithms
- ✓ "Plug-and-Play" path planning module
- ✓ 2 path planning algorithms swapping – lawnmower & perimeter spiral

☐ Enhances system flexibility and efficiency

If you'd like to know more, you can search online later for this project: <https://github.com/ChuanXiixiii/Group-Project-in-Aerial-Robotics>

Path Planning Algorithms – Lawnmower

Core Idea:

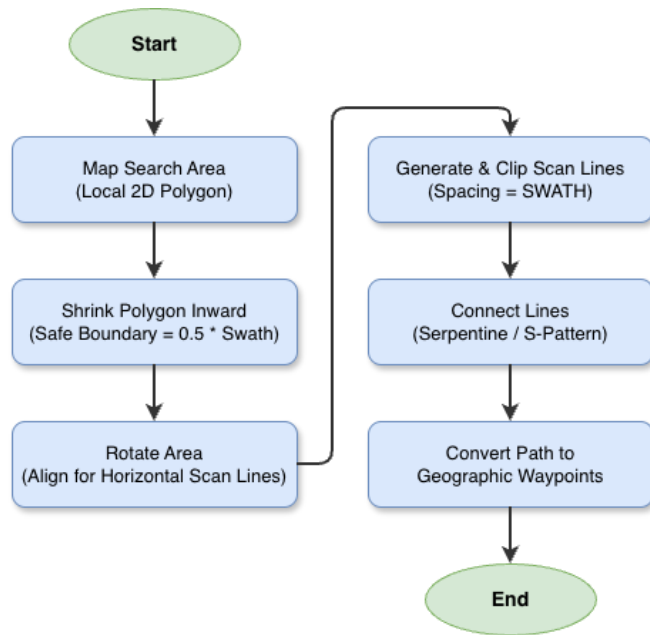
Prioritise Coverage Completeness

+

Regularised Scanning

+

Minimise Transition Costs



Path Planning Algorithms – Perimeter Spiral

Core Idea:

Outer-to-inner rings

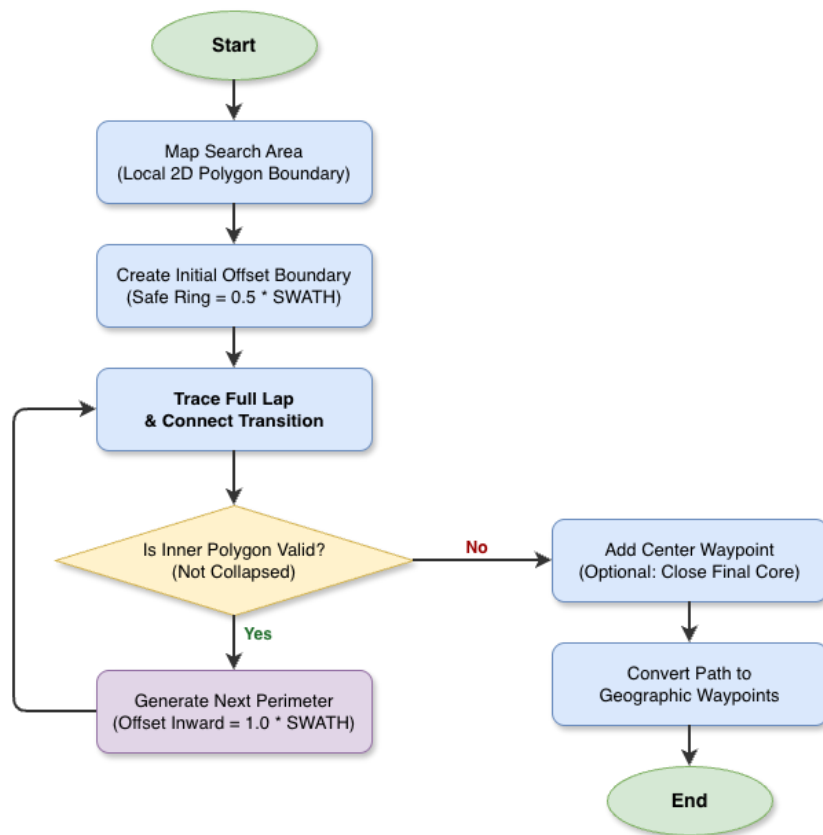
+

Swath-based inward offset

+

Perimeter lap repetition

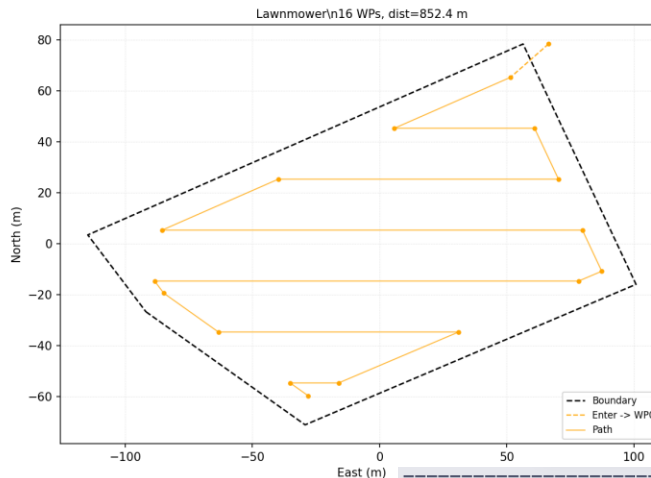
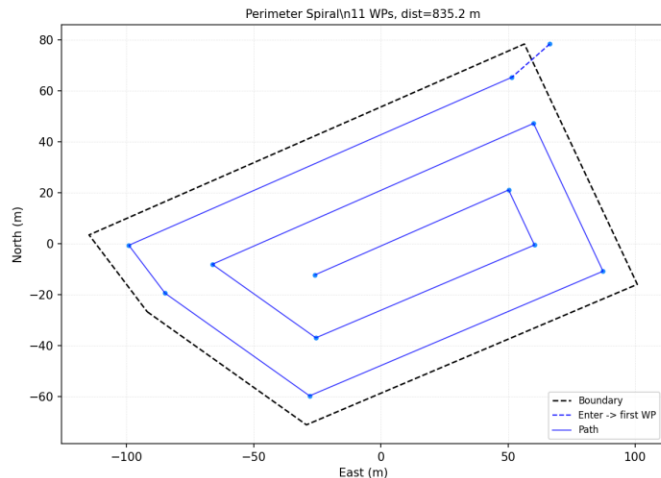
Compared with *lawnmower*, it usually has fewer sharp back-and-forth turns, but path efficiency depends strongly on polygon shape.



Performance of 2 Path Planning Algorithms

Use Spacing = 20.0 m as example,

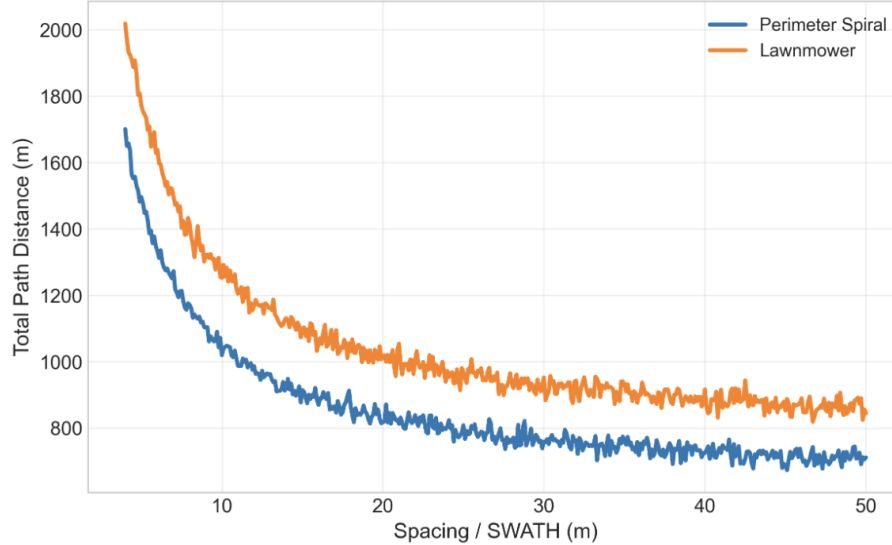
Path Planning Comparison — Fenswood Farm
 HALF_SWATH=10.0 m, SWATH=20.0 m



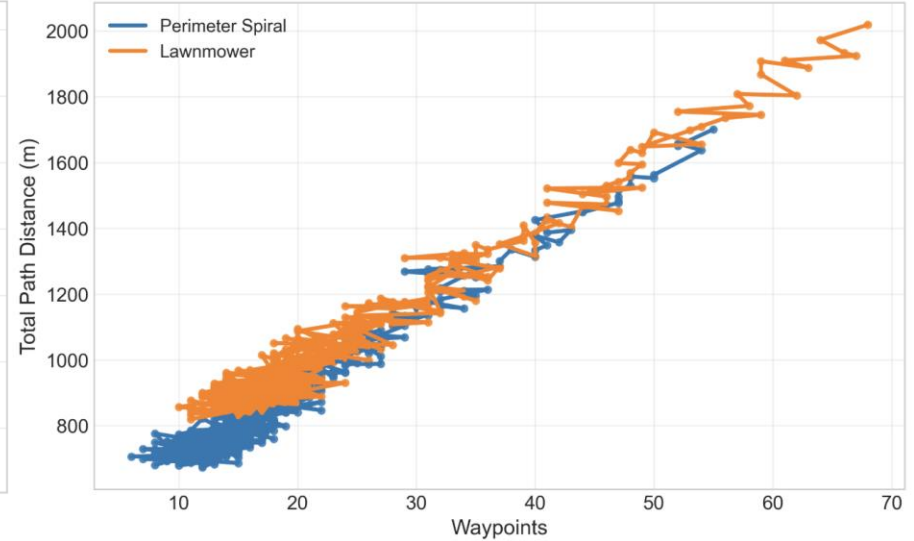
ALGORITHM COMPARISON

Metric	Perimeter Spiral	Lawnmower
Waypoints (excl. enter)	11	16
Total distance (m)	835.2	852.4
HALF_SWATH (m)	10.0	10.0
SWATH / line spacing (m)	20.0	20.0

Path Planning Simulation



Distance vs. Spacing



Distance vs. Waypoints

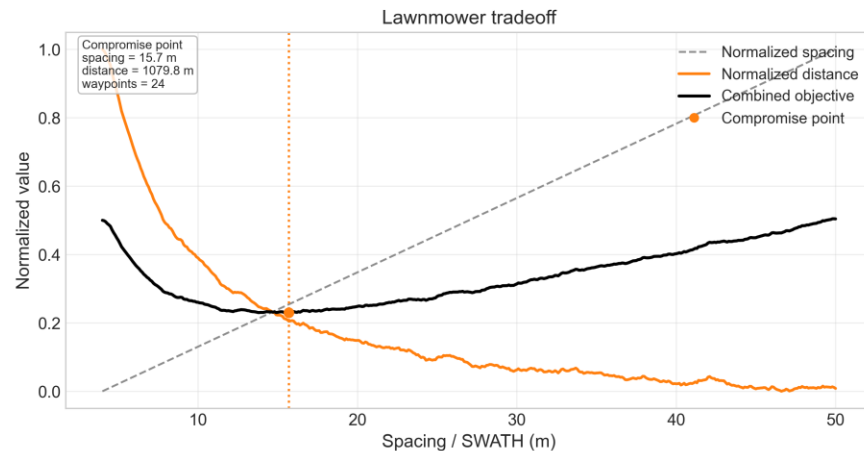
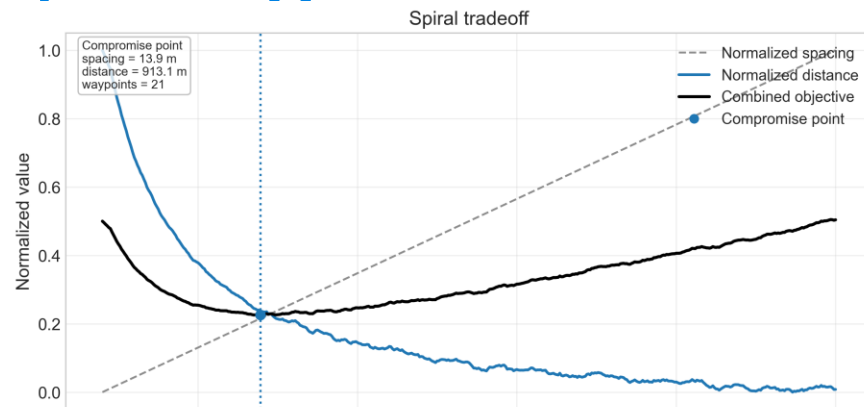
Compromise Point: Spacing Selection

Equal-weight tradeoff between spacing and flight distance:

$$\text{score} = 0.5 \text{norm}(\text{spacing}) + 0.5 \text{norm}(\text{distance})$$

Result: *at the compromise point

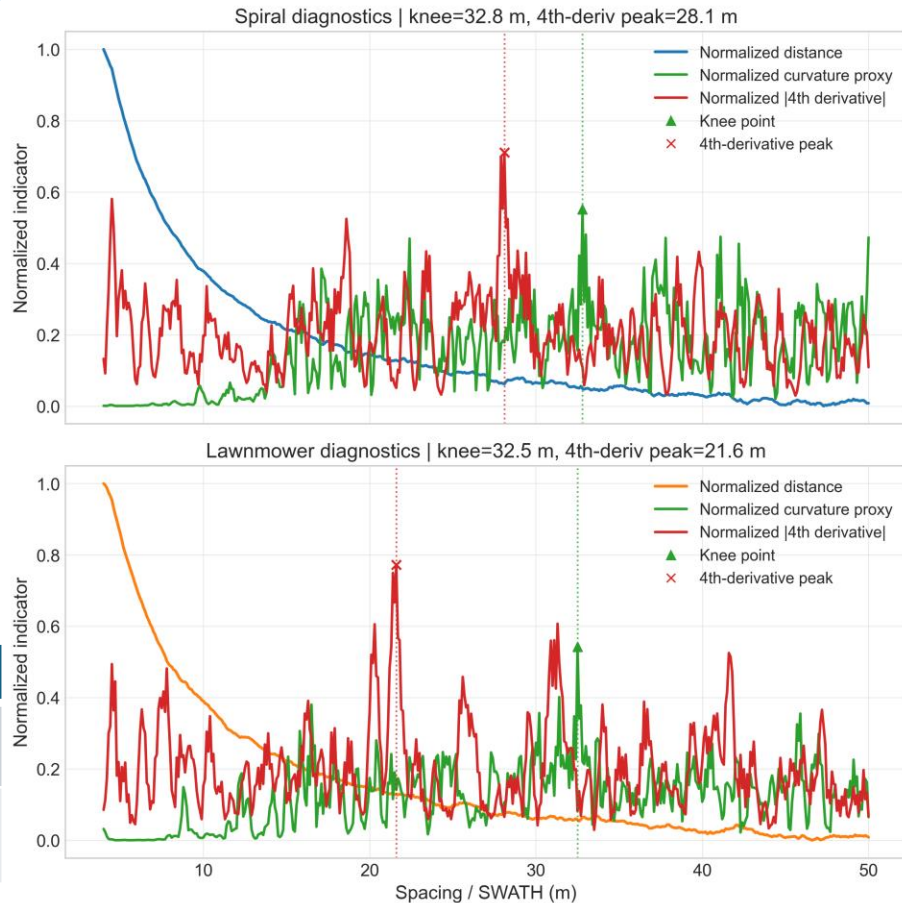
Algorithm	Spacing(m)	Distance(m)	Waypoints
Lawnmover	15.7	1079.8	24
Perimeter spiral	13.9	913.1	21



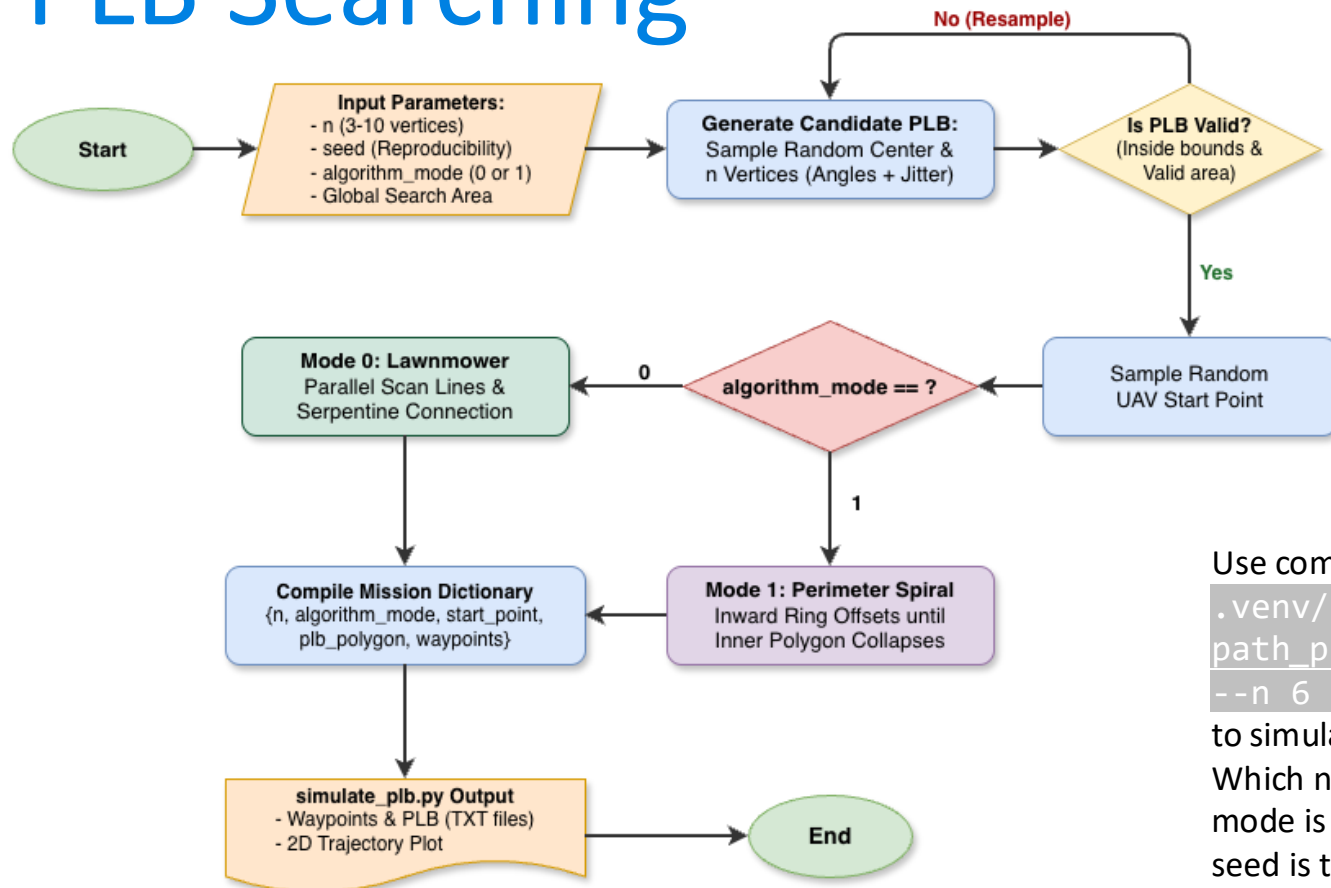
Knee Point + 4th Derivative Peak

- ❑ **Knee point** indicates where marginal benefit starts to diminish (>37.5m in spiral and >32.8m in lawnmover).
- ❑ **4th-derivative peak** indicates a strong shape-change region in the performance curve, but it's easily influenced by noise.
- ❑ These two points are diagnostic references, not standalone final decisions.

Algorithm	knee(m)	4 th -deriv peak(m)
Lawnmover	32.5	21.6
Perimeter spiral	13.9	28.1



PLB Searching



Use command

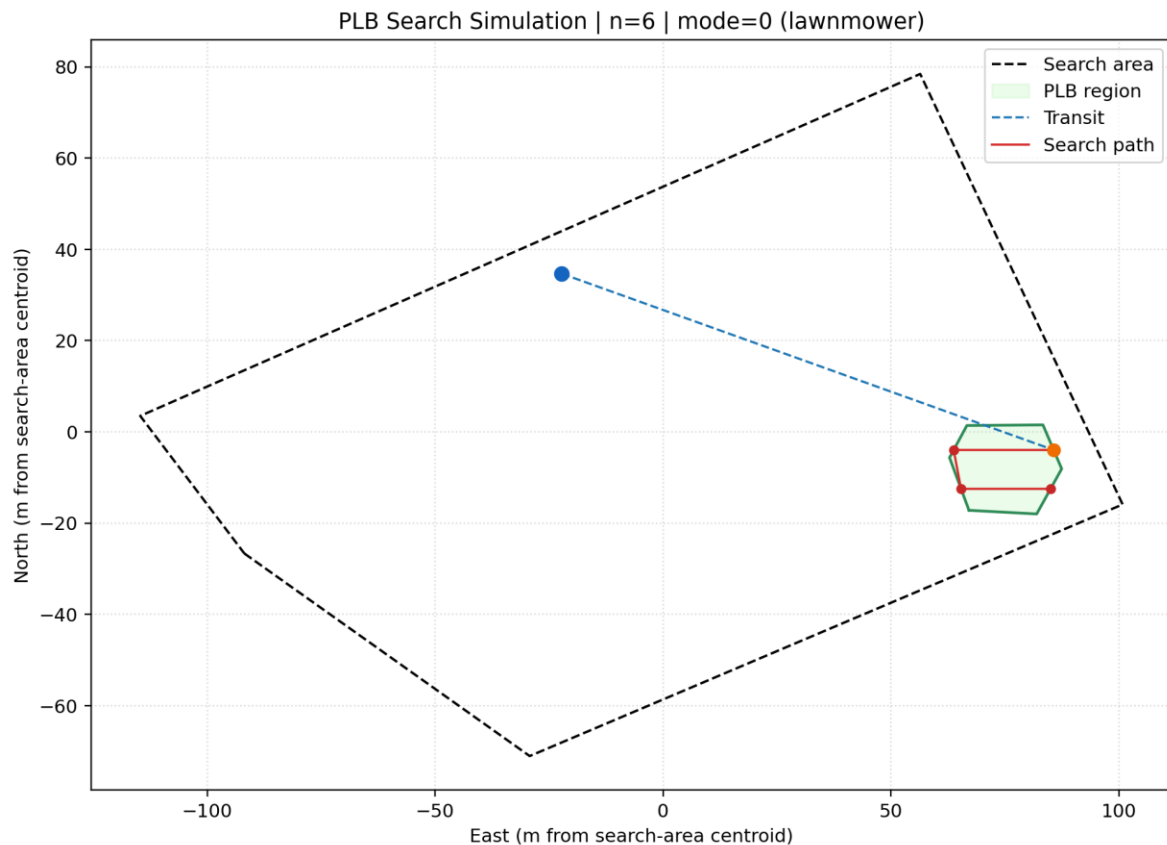
```
.venv/bin/python  
path_planner/simulate_plb.py  
--n 6 --mode 0 --seed 323
```

to simulate.

Which **n** is number of sides of PLB;
mode is the path planning method;
seed is the random number used to
generate the polygon

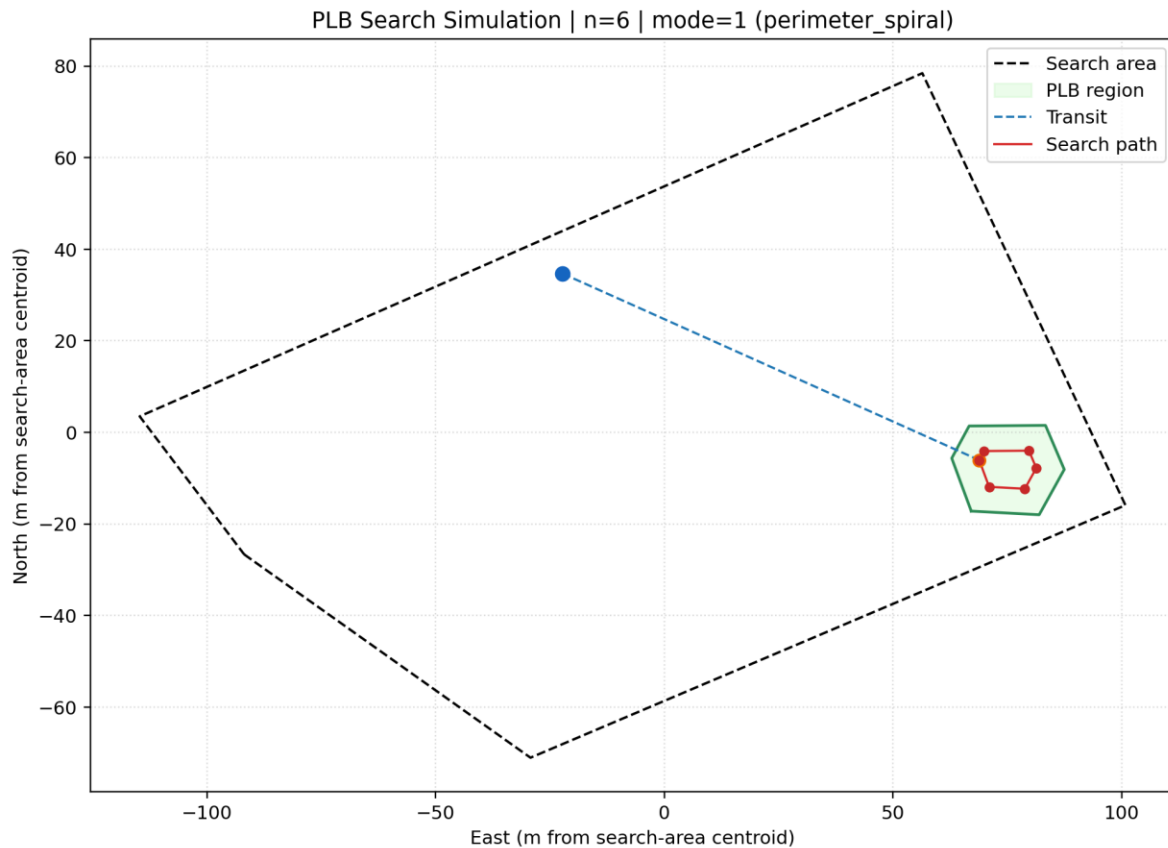
Performance of PLB Searching Simulation

□ Use --n 6 --mode 0 --seed 1224 as example,



Performance of PLB Searching Simulation

□ Use --n 6 --mode 1 --seed 1224 as example,



Team Formation and Operation

Meetings and Communication

- in-person meetings (60–120 minutes)
- Online check-ins when needed
- A balance between structured discussions and quick updates

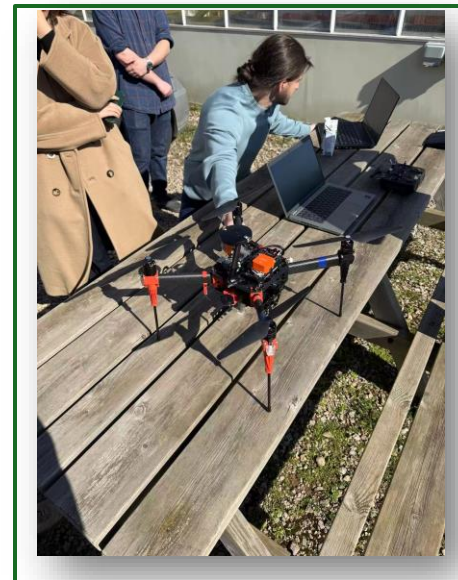
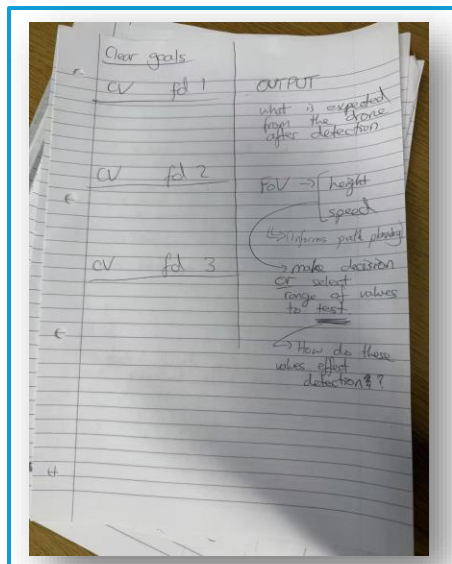
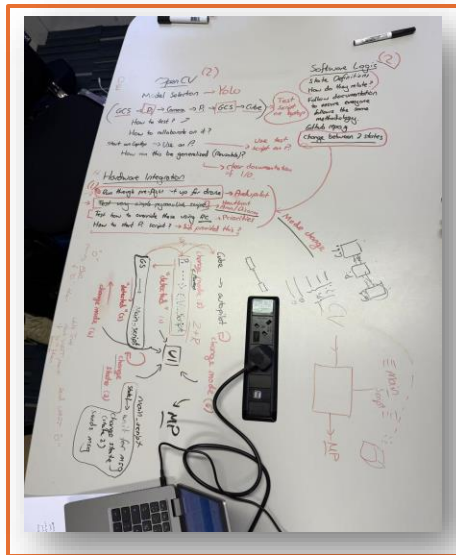
Task and Workflow Management

- Tasks organised based on system modules
 - Parallel development across components
 - Continuous integration and testing
- Strong communication ensured alignment

Leadership and Team Dynamics

Shared leadership model

- Each member responsible for their module
 - Decisions made collaboratively
- Strong coordination between vision and planning



Team Structure and Collaboration



Our Team



Dmytro



Edward



Robin



Herish



Ziyan

Our Team



Dmytro



Edward



Robin



Herish



Ziyang

- ☐ Accelerated SITL Development
- ☐ Computer Vision Model Implementation

Our Team



Dmytro



Edward



Robin



Herish



Ziyan

- ☐ Synthetic CV Training Data
- ☐ CV Model Training

Our Team



Dmytro



Edward



Robin



Herish



Ziyang

- ☐ Safety-Critical State Machine
- ☐ Group Software Integration

Our Team



Dmytro



Edward



Robin



Herish



Ziyan

- ☐ PLB Replanning Algorithm
- ☐ Team Co-ordinator

Our Team



Dmytro



Edward



Robin



Herish



Ziyang

☐ Path Planning Optimisation

☐ ...

